

Problem Statement

Given an array of 1s and 0s this has all 1s first followed by all 0s. Aim is to find the number of 0s. Write a program using Divide and Conquer to Count the number of zeroes in the given array.

Input Format

First Line Contains Integer m – Size of array

Next m lines Contains m numbers – Elements of an array

Output Format

First Line Contains Integer – Number of zeroes present in the given array.

Answer: (penalty regime: 0 %)

```
1  #include <stdio.h>
2
3  // Divide and Conquer: find the index of the first 0
4  int findFirstZero(int a[], int low, int high) {
5      if (low > high)
6          return -1;
7
8      int mid = low + (high - low) / 2;
9
10     // Check whether a[mid] is the first zero
11     if (a[mid] == 0 && (mid == 0 || a[mid - 1] == 1))
12         return mid;
13
14     if (a[mid] == 1)
15         return findFirstZero(a, mid + 1, high);
16     else
17         return findFirstZero(a, low, mid - 1);
18 }
19
20 int main() {
21     int m;
22     scanf("%d", &m);
23
24     int a[m];
25
26     for (int i = 0; i < m; i++) {
27         scanf("%d", &a[i]);
28     }
29
30     int firstZero = findFirstZero(a, 0, m - 1);
31
32     if (firstZero == -1)
33         printf("0\n");
34     else
35         printf("%d\n", m - firstZero);
36
37     return 0;
38 }
```

	Input	Expected	Got	
✓	5 1 1 1 0 0	2	2	✓
✓	10 1 1 1 1 1 1 1 1 1 1	0	0	✓
✓	8 0 0 0 0 0 0 0 0	8	8	✓

✓	17	2	2	✓
	1			
	1			
	1			
	1			
	1			
	1			
	1			
	1			
	1			
	1			
	1			
	1			
	1			
	1			
	1			
	0			
	0			

Passed all tests! ✓